

1 Working with Matrices and Vectors 3

1.1 Notation for Scalars, Vectors, and Matrices 3

1.2 Matrix and Vector Operations . . . 3

1.3 Determinant 17

1.4 Orthogonal and Idempotent Matrices 20

1.5 Linear Combinations and Column Spaces 21

1.6 Linear Independence 23

1.7 Rank 25

1.8 Inverse 28

1.9 Trace 33

1.10 Eigenvalues and Eigenvectors . . . 36

1.11 Quadratic Form 39

1.11.1 Helmert Matrix 42

1.11.2 Symmetric Matrices 44

1.11.3 Positive Definiteness 45

1.12 Spectral Decomposition 48

1.13 Random Vectors: 51

1.13.1 Mean vectors: 51

1.13.2 Covariance matrix: 52

1 Working with Matrices and Vectors

1.1 Notation for Scalars, Vectors, and Matrices

Lowercase letters $\Rightarrow$ scalars: x ; c ; σ .

Boldface, lowercase letters $\Rightarrow$ vectors: $\mathbf{x}$; $\mathbf{y}$; $\boldsymbol{\beta}$.

Boldface, uppercase letters $\Rightarrow$ matrices: $\mathbf{A}$; $\mathbf{X}$; $\boldsymbol{\Sigma}$.

1.2 Matrix and Vector Operations

Definition 1.
A column of real numbers is called a **vector**.

Example 1.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \mathbf{a} = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Since $\mathbf{y}$ has n elements it is said to have **order** (or dimension) n .

Definition 2.

A rectangular array of elements with m rows and k columns is called an $m \times k$ **matrix**.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix}$$

This matrix is said to be of **order** (or dimension) $m \times k$, where

- m is the **row** order (dimension)
- k is the **column** order (dimension)

Example 2.

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 \\ 0 & 4 & 5 \end{bmatrix} \quad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

R-code:

```
A = matrix(c(1,3,-2,0,4,5), 2,3, byrow = T)
A
I = diag(3)
I
B = matrix(c(1,3,2,6), 2,2, byrow = T)
B
```

Definition 3. Matrix addition

If $\mathbf{A}$ and $\mathbf{B}$ are both $m \times k$ matrices, then

$$\mathbf{C} = \mathbf{A} + \mathbf{B}$$

$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mk} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1k} \\ b_{21} & b_{22} & \cdots & b_{2k} \\ \vdots & \vdots & & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mk} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1k} + b_{1k} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2k} + b_{2k} \\ \vdots & \vdots & & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \cdots & a_{mk} + b_{mk} \end{bmatrix}$$

Notation:

$$C_{m \times k} = \{c_{ij}\} \text{ where } c_{ij} = a_{ij} + b_{ij}$$

Definition 4. Matrix subtraction

If $\mathbf{A}$ and $\mathbf{B}$ are $m \times k$ matrices, then $\mathbf{C} = \mathbf{A} - \mathbf{B}$ is defined by

$$\mathbf{C} = \{c_{ij}\} \text{ where } c_{ij} = a_{ij} - b_{ij} .$$

Example 3.

$$\begin{bmatrix} 3 & 6 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 7 & -4 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 2 \\ -1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} 1 & -1 \\ 2 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ -1 & 1 \\ 0 & -1 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,6,2,1), 2,2,byrow=T)
B = matrix(c(7,-4,-3,2), 2,2,byrow=T)
C = A+B
D = matrix(c(1,-1,1,1,0), 3,2,byrow=T)
E = matrix(c(1,-1,2,0,1,1), 3,2,byrow=T)
F = D-E
```

Definition 5. Scalar multiplication

Let a be a scalar and $\mathbf{B} = \{b_{ij}\}$ be an $m \times k$ matrix, then

$$a\mathbf{B} = \mathbf{B}a = \{ab_{ij}\}$$

Example 4.

$$2 \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & -2 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 6 \\ 0 & 8 & -4 \end{bmatrix}$$

```
R-Code:
A = matrix(c(2,-1,3,0,4,2), 2,3,byrow=T)
B= 2*A
```

Definition 6. Transpose

The transpose of the $m \times k$ matrix $\mathbf{A} = \{a_{ij}\}$ is the $k \times m$ matrix with elements $\{a_{ji}\}$. The transpose of $\mathbf{A}$ is denoted by $\mathbf{A}^T$ (or $\mathbf{A}'$).

Example 5.

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 3 & 0 \\ -2 & 6 \end{bmatrix} \quad \mathbf{A}^T = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 0 & 6 \end{bmatrix}$$

```
R-code:
A = matrix(c(1,4,3,0,-2,6), 3,2,byrow=T)
AT = t(A)
AT
```

Definition 7. If a matrix has the same number of rows and columns it is called a **square matrix**.

$$\mathbf{A}_{k \times k} = \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$$

is said to have order (or dimension) k .

Definition 8. A square matrix $\mathbf{A} = \{a_{ij}\}$ is **symmetric** if $\mathbf{A} = \mathbf{A}^T$, that is, if $a_{ij} = a_{ji}$ for all (i, j) .

Example 6.

$$\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 5 & 0 & -2 \\ 2 & 0 & 3 & -1 \\ 1 & -2 & -1 & 2 \end{bmatrix}$$

Definition 9. Inner product (crossproduct) of two vectors of order n

$$\mathbf{a}^T \mathbf{y} = [a_1, a_2, \dots, a_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$= [a_1 y_1 + a_2 y_2 + \cdots + a_n y_n]$$

$$= \sum_{j=1}^n a_j y_j$$

Note that $\mathbf{a}^T \mathbf{y} = \mathbf{y}^T \mathbf{a}$

```
R-codes:
a = c(1, 7, -6, 4)
y = c(2,-2,1,5)
aTy1 = t(a)%*%y
aTy2 = a%*%y
aTy3 = crossprod(a,y)
```

Definition 10. Matrix multiplication

The product of an $n \times k$ matrix $\mathbf{A}$ and a $k \times m$ matrix $\mathbf{B}$ is the $n \times m$ matrix $\mathbf{C} = \{c_{ij}\}$ with elements

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ik}b_{kj}$$

Example 7.

$$\mathbf{A} = \begin{bmatrix} 3 & 0 & -2 \\ 1 & -1 & 4 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \quad \mathbf{C} = \mathbf{AB} = \begin{bmatrix} 1 & -3 \\ 4 & 11 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,0,-2,1,-1,4), 2,3,byrow=T)
B = matrix(c(1,1,1,2,1,3), 3,2,byrow=T)
C = A%B
```

Definition 11. Elementwise multiplication of two matrices

$$\mathbf{A} \# \mathbf{B} = \begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \vdots \\ a_{k1} & \cdots & a_{km} \end{bmatrix} \# \begin{bmatrix} b_{11} & \cdots & b_{1m} \\ \vdots & & \vdots \\ b_{k1} & \cdots & b_{km} \end{bmatrix} \\ = \begin{bmatrix} a_{11}b_{11} & \cdots & a_{1m}b_{1m} \\ \vdots & & \vdots \\ a_{k1}b_{k1} & \cdots & a_{km}b_{km} \end{bmatrix}$$

Example 8.

$$\begin{bmatrix} 3 & 1 \\ 2 & 4 \\ 0 & 6 \end{bmatrix} \# \begin{bmatrix} 1 & -5 \\ -3 & 4 \\ -2 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -5 \\ -6 & 16 \\ 0 & 12 \end{bmatrix}$$

R-codes:

```
A = matrix(c(3,1,2,4,0,6), 3,2,byrow=T)
B = matrix(c(1,-5,-3,4,-2,2), 3,2,byrow=T)
C = A*B
```

Definition 12. Kronecker product of two matrices

$$\mathbf{A}_{k \times m} \otimes \mathbf{B}_{n \times s} = \begin{bmatrix} a_{11}\mathbf{B} & a_{12}\mathbf{B} & \cdots & a_{1m}\mathbf{B} \\ a_{21}\mathbf{B} & a_{22}\mathbf{B} & \cdots & a_{2m}\mathbf{B} \\ \vdots & \vdots & & \vdots \\ a_{k1}\mathbf{B} & a_{k2}\mathbf{B} & \cdots & a_{km}\mathbf{B} \end{bmatrix}$$

Example 9.

$$\begin{bmatrix} 2 & 4 \\ 0 & -2 \\ 3 & -1 \end{bmatrix} \otimes \begin{bmatrix} 5 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 10 & 6 & 20 & 12 \\ 4 & 2 & 8 & 4 \\ 0 & 0 & -10 & -6 \\ 0 & 0 & -4 & -2 \\ 15 & 9 & -5 & -3 \\ 6 & 3 & -2 & -1 \end{bmatrix}$$

$$\mathbf{a} \otimes \mathbf{y} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \otimes \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_1 y_1 \\ a_1 y_2 \\ a_2 y_1 \\ a_2 y_2 \\ a_3 y_1 \\ a_3 y_2 \end{bmatrix}$$

R-codes:

```
A = matrix(c(2,4,0,-2,3,-1),ncol=2,byrow=T)
B = matrix(c(5,3,2,1),2,2,byrow=T)
C = kronecker(A,B)
```

1.3 Determinant

Definition 13. The **determinant** of an $n \times n$ matrix $\mathbf{A}$ is

$$|\mathbf{A}| = \sum_{j=1}^n a_{ij}(-1)^{i+j}|M_{ij}| \text{ for any row } i$$

or

$$|\mathbf{A}| = \sum_{i=1}^n a_{ij}(-1)^{i+j}|M_{ij}| \text{ for any column } j$$

where M_{ij} is the “minor” for a_{ij} obtained by deleting the i^{th} row and j^{th} column from A .

Example 10.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$|\mathbf{A}| = a_{11}(-1)^{1+1}|a_{22}| + a_{12}(-1)^{1+2}|a_{21}|$$

$$\text{then } \begin{vmatrix} 7 & 2 \\ 4 & 5 \end{vmatrix} =$$

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Example 11.

$$\begin{aligned} |\mathbf{A}| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \\ &= a_{11}(-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + a_{12}(-1)^{1+2} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} \\ &\quad + a_{13}(-1)^{1+3} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \end{aligned}$$

$$\text{then } \begin{vmatrix} 1 & 1 & 3 \\ 4 & 3 & 6 \\ 7 & 5 & 9 \end{vmatrix} =$$

R-codes:

```
A = matrix(c(1,1,3,4,3,6,7,5c',9),3,3,byrow=T)
> detA = det(A)
> detA
```

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Properties of determinants

- $|\mathbf{A}^T| = |\mathbf{A}|$
- $|\mathbf{A}|$ = product of the eigenvalues of $\mathbf{A}$
- $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$ when $\mathbf{A}$ and $\mathbf{B}$ are square matrices of the same order.
- $\begin{vmatrix} \mathbf{P} & \mathbf{0} \\ \mathbf{X} & \mathbf{Q} \end{vmatrix} = |\mathbf{P}||\mathbf{Q}|$ when $\mathbf{P}$ and $\mathbf{Q}$ are square matrices of the same order and $\mathbf{0}$ is a matrix of zeros.
- $|\mathbf{AB}| = |\mathbf{BA}|$ when the matrix product is defined
- $|c\mathbf{A}| = c^k|\mathbf{A}|$ when c is a scalar and $\mathbf{A}$ is a $k \times k$ matrix

1.4 Orthogonal and Idempotent Matrices

Definition 14. A square matrix $\mathbf{A}$ is said to be **orthogonal** if

$$\mathbf{AA}^T = \mathbf{A}^T\mathbf{A} = \mathbf{I} \quad (\text{then } \mathbf{A}^{-1} = \mathbf{A}^T)$$

Definition 15. A square matrix $\mathbf{P}$ is **idempotent** if $\mathbf{PP} = \mathbf{P}$

Example 12. (Orthogonal Matrix)

$$\mathbf{A} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

In each case the columns of $\mathbf{A}$ are coefficients for orthogonal contrasts.

Example 13. (Idempotent Matrix)

$$\mathbf{P} = \begin{bmatrix} \frac{5}{6} & \frac{2}{6} & -\frac{1}{6} \\ \frac{2}{6} & \frac{2}{6} & \frac{2}{6} \\ -\frac{1}{6} & \frac{2}{6} & \frac{5}{6} \end{bmatrix}$$

1.5 Linear Combinations and Column Spaces

$\mathbf{A}\mathbf{b}$ is a linear combination of the columns of an $m \times n$ of matrix $\mathbf{A}$.

$$\mathbf{A}\mathbf{b} = [\mathbf{a}_1, \dots, \mathbf{a}_n] \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} = b_1\mathbf{a}_1 + \dots + b_n\mathbf{a}_n$$

The set of all possible linear combinations of the columns of $\mathbf{A}$ is called the column space of $\mathbf{A}$ and is written as

$$\mathcal{C}(\mathbf{A}) = \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbf{R}^n\}$$

Note that $\mathcal{C}(\mathbf{A}) \subseteq \mathbf{R}^m$.

Example 14.

Suppose $\mathbf{X}$ is an $n \times p$ matrix and $\mathbf{B}$ is a $p \times p$ non-singular matrix. Prove that $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}\mathbf{B}^{-1})$.

1.6 Linear Independence

Definition 16. A set of n -dimensional vectors $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_k$ are **linearly independent** if there is no set of scalars $a_1, a_2, \dots, a_k$ such that

$$\mathbf{0} = \sum_{j=1}^k a_j \mathbf{y}_j$$

and at least one a_j is non-zero.

Example 15. Show that

$$\mathbf{y}_1 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

are linearly independent.

Example 16. Show that

$$\mathbf{y}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

are not linearly independent.

1.7 Rank

Definition 17. The **row rank** of a matrix is the number of linearly independent rows, where each row is considered as a vector.

Definition 18. The **column rank** of a matrix is the number of linearly independent columns, with each column considered as a vector.

Example 17. Show that the row and column rank of

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 5 & -1 \\ 0 & 1 & -1 \end{bmatrix}$$

is 2.

R-codes:

```
A = matrix(c(1,1, 1,2,5,-1,0,1,-1),3,3,byrow=T)
rA = qr(A)$rank
```

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Result 1. The row rank and the column rank of a matrix are equal.

Definition 19. The **rank** of a matrix is either the row rank or the column rank of the matrix.

Definition 20. A square matrix $\mathbf{A}_{k \times k}$ is **non-singular** if its rank is equal to the number of rows (or columns).

This is equivalent to the condition

$$\mathbf{A}_{k \times k} \mathbf{b}_{k \times 1} = \mathbf{0}_{k \times 1} \text{ only when } \mathbf{b} = \mathbf{0}$$

A matrix that fails to be nonsingular is called **singular**.

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Result 2. If $\mathbf{B}_{n \times n}$ is non-singular and $\mathbf{A}_{n \times m}$, then

$$\text{rank}(\mathbf{BA}) = \text{rank}(\mathbf{A}).$$

Result 3. If $\mathbf{B}$ and $\mathbf{C}$ are non-singular matrices and products with $\mathbf{A}$ are defined, then

$$\text{rank}(\mathbf{BA}) = \text{rank}(\mathbf{AC}) = \text{rank}(\mathbf{A}).$$

Result 4. $\text{rank}(\mathbf{A}^T \mathbf{A}) = \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^T)$.

1.8 Inverse

Definition 21. The **identity matrix**, denoted by $\mathbf{I}$, is a $k \times k$ matrix of the form

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Definition 22. The **inverse** of a square, non-singular matrix $\mathbf{A}$ is the matrix, denoted by $\mathbf{A}^{-1}$, such that

$$\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

R-codes:

```
I3 = diag(rep(1,3))
I3
W = matrix(c(1,2,3,4,5,6,7,8,10),3,3,byrow=T)
Winv = solve(W)
Winv
```

Result 5.

(i) The inverse of $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

(ii) In general, the (i, j) element of $\mathbf{A}^{-1}$ is

$$\frac{(-1)^{i+j} |\mathbf{A}_{ji}|}{|\mathbf{A}|}$$

where $\mathbf{A}_{ji}$ is the matrix obtained by deleting the j -th row and i -th column of $\mathbf{A}$.

Result 6. For a $k \times k$ matrix $\mathbf{A}$, the following are equivalent:

- (i) $\mathbf{A}$ is nonsingular
- (ii) $|\mathbf{A}| \neq 0$
- (iii) $\mathbf{A}^{-1}$ exists

Result 7. For $k \times k$ nonsingular matrices $\mathbf{A}$ and $\mathbf{B}$

- (i) $(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$
- (ii) $(\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
- (iii) $|\mathbf{A}^{-1}| = 1/|\mathbf{A}|$
- (iv) $\mathbf{A}^{-1}$ is unique and nonsingular
- (v) $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- (vi) If $\mathbf{A}$ is symmetric, then $\mathbf{A}^{-1}$ is symmetric

Result 8. Inverse of a Diagonal Matrix

$$\begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{kk} \end{bmatrix}^{-1} = \begin{bmatrix} 1/a_{11} & & & \\ & 1/a_{22} & & \\ & & \ddots & \\ & & & 1/a_{kk} \end{bmatrix}$$

Result 9.

If $\mathbf{B}$ is a $k \times k$ non-singular matrix and $\mathbf{B} + \mathbf{c}\mathbf{c}^T$ is non-singular, then

$$(\mathbf{B} + \mathbf{c}\mathbf{c}^T)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{c}\mathbf{c}^T\mathbf{B}^{-1}}{1 + \mathbf{c}^T\mathbf{B}^{-1}\mathbf{c}}$$

Result 10.

Let $\mathbf{I}_n$ be an $n \times n$ identity matrix and let $\mathbf{J}_n = \mathbf{1}\mathbf{1}^T$ be an $n \times n$ matrix where each element is one, then

$$(a\mathbf{I}_n + b\mathbf{J}_n)^{-1} = \frac{1}{a} \left(\mathbf{I}_n - \frac{b}{a + nb}\mathbf{J}_n \right)$$

Example 18.

Suppose $\mathbf{Z} = \mathbf{1}_{6 \times 1}$, $\mathbf{G} = 25$, $\mathbf{R} = 49 \mathbf{I}_{6 \times 6}$. If $\Sigma = \mathbf{Z}\mathbf{G}\mathbf{Z}^T + \mathbf{R}$, find Σ^{-1} .

1.9 Trace

Definition 23. The **trace** of a $k \times k$ matrix $\mathbf{A} = \{a_{ij}\}$ is the sum of the diagonal elements:

$$tr(\mathbf{A}) = \sum_{j=1}^k a_{jj}$$

R-codes:

```
W = matrix(c(1,2,3,4,5,6,7,8,10),
            3,3,byrow=T)
trW = sum(diag(W))
trW
```

R-codes:

```
W = {1 2 3, 4 5 6, 7 8 10};
trW1 = trace(W);
trW2 = sum(diag(W));
print W trW1 trW2;
```

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Result 11. Let $\mathbf{A}$ and $\mathbf{B}$ denote $k \times k$ matrices and let c be a scalar. Then,

- (i) $tr(c\mathbf{A}) = c\,tr(\mathbf{A})$
- (ii) $tr(\mathbf{A} + \mathbf{B}) = tr(\mathbf{A}) + tr(\mathbf{B})$
- (iii) $tr(\mathbf{AB}) = tr(\mathbf{BA})$
- (iv) $tr(\mathbf{B}^{-1}\mathbf{AB}) = tr(\mathbf{A})$
- (v) $tr(\mathbf{A}\mathbf{A}^T) = \sum_{i=1}^k \sum_{j=1}^k a_{ij}^2$

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Example 19.

For $\mathbf{A} = \mathbf{I}_{n \times n} - \frac{1}{n}\mathbf{i}\mathbf{i}^T$ where $\mathbf{i}^T = [1 \ 1 \ \cdots \ 1]_{1 \times n}$.

- (a) Show that $\mathbf{A}$ is idempotent.
- (b) Find $tr(\mathbf{A})$.
- (c) Interpret the result of $\mathbf{A}\mathbf{y}$ where $\mathbf{y}$ is $n \times 1$.

1.10 Eigenvalues and Eigenvectors

Definition 24. For a $k \times k$ matrix $\mathbf{A}$, the scalars $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$ satisfying the polynomial equation

$$|\mathbf{A} - \lambda \mathbf{I}| = 0$$

are called the eigenvalues (or characteristic roots) of $\mathbf{A}$.

Definition 25. Corresponding to any eigenvalue λ_i is an eigenvector (or characteristic vector) $\mathbf{u}_i \neq \mathbf{0}$ satisfying

$$\mathbf{A}\mathbf{u}_i = \lambda_i \mathbf{u}_i.$$

Comment: Eigenvectors are not unique

- (i) If $\mathbf{u}_i$ is an eigenvector for λ_i , then $c\mathbf{u}_i$ is also an eigenvector for any scalar $c \neq 0$.

(ii) We will adopt the following conventions (for real symmetric matrices)

$$\mathbf{u}_i^T \mathbf{u}_i = 1 \quad \text{for all } i = 1, \dots, k$$

$$\mathbf{u}_i^T \mathbf{u}_j = 0 \quad \text{for all } i \neq j$$

(iii) Even with (ii), eigenvectors are not unique

- If $\mathbf{u}_i$ is an eigenvector satisfying (ii), then $-\mathbf{u}_i$ is also an eigenvector satisfying (ii).
- If $\lambda_i = \lambda_j$ then there are an infinite number of choices for $\mathbf{u}_i$ and $\mathbf{u}_j$.

Result 12. For a $k \times k$ symmetric matrix $\mathbf{A}$ with elements that are real numbers

- (i) every eigenvalue of $\mathbf{A}$ is a real number
- (ii) $\text{rank}(\mathbf{A}) = \text{number of non-zero eigenvalues}$
- (iii) if $\mathbf{A}$ is non-negative definite, then $\lambda_i \geq 0$ for all $i = 1, 2, \dots, k$
- (iv) if $\mathbf{A}$ is positive definite then $\lambda_i > 0$ for all $i = 1, 2, \dots, k$
- (v) $\text{trace}(\mathbf{A}) = \sum_{i=1}^k a_{ii} = \sum_{i=1}^k \lambda_i$
- (vi) $|\mathbf{A}| = \prod_{i=1}^k \lambda_i$
- (vii) if $\mathbf{A}$ is idempotent ($\mathbf{A}\mathbf{A} = \mathbf{A}$), then the eigenvalues are either zero or one.

1.11 Quadratic Form

Definition 26.

Let $\mathbf{A}$ be a $k \times k$ matrix and let $\mathbf{y}$ be a vector of order k , then

$$\mathbf{y}^T \mathbf{A} \mathbf{y} = \sum_{i=1}^k \sum_{j=1}^k y_i y_j a_{ij}$$

is called a **quadratic form**.

Suppose $\mathbf{y}_{n \times 1}$ is a vector of n observations. Then $\mathbf{y}^T \mathbf{y} = \sum_{i=1}^n y_i^2$ is the total sum of squares of the observations. Let $\mathbf{P}$ be an orthogonal matrix

$$\mathbf{P}\mathbf{P}^T = \mathbf{P}^T\mathbf{P} = \mathbf{I}$$

and partition $\mathbf{P}$ row wise into k sub-matrices $\mathbf{P}_i$, of order $n_i \times n$, for $i = 1, 2, \dots, k$, with $\sum_{i=1}^k n_i = n$; i.e.

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \\ \vdots \\ \mathbf{P}_k \end{bmatrix} \quad \text{and} \quad \mathbf{P}^T = [\mathbf{P}_1^T \quad \mathbf{P}_2^T \quad \dots \quad \mathbf{P}_k^T].$$

Then

$$\mathbf{y}^T \mathbf{y} = \mathbf{y}^T \mathbf{I} \mathbf{y} = \mathbf{y}^T \mathbf{P}^T \mathbf{P} \mathbf{y} = \sum_{i=1}^k \mathbf{y}^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{y}.$$

In this way $\mathbf{y}^T \mathbf{y}$ is partition into k sums of squares

$$\mathbf{y}^T \mathbf{P}_i^T \mathbf{P}_i \mathbf{y} \quad \text{for } i = 1, \dots, k$$

each of these sums of squares corresponds to the lines in an analysis of variance, having $\mathbf{y}^T \mathbf{y}$ as the total sums of squares.

Example 20.

Corresponding to a vector of 4 observations consider

$$\mathbf{P} = \begin{bmatrix} \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} & \frac{1}{\sqrt{4}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{12}} & \frac{-3}{\sqrt{12}} \end{bmatrix} = \begin{bmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \end{bmatrix}$$

Show that $\mathbf{P}$ is orthogonal and find the two partition sums of squares.

1.11.1 Helmert Matrix

The matrix $\mathbf{P}$ in Example 20 is a fourth-order Helmert matrix. Writing

$$\mathbf{H}_{n \times n} = \begin{bmatrix} \mathbf{h}^T \\ \mathbf{H}_0 \end{bmatrix} \begin{matrix} 1 \times n \\ (n-1) \times n \end{matrix}$$

for a Helmert matrix of order n ,

$$\mathbf{h}^T = \text{first row of } \mathbf{H}_{n \times n} = \frac{1}{\sqrt{n}} \mathbf{1}_n^T$$

and

$$\mathbf{H}_0 = \text{last } (n-1) \text{ rows of } \mathbf{H}_{n \times n}$$

with $\mathbf{H}_0$ having its r th row as

$$\left[\frac{1}{\sqrt{r(r+1)}} \mathbf{1}_r^T \quad \frac{-r}{\sqrt{r(r+1)}} \mathbf{0}_{(n-r-1) \times 1} \right]$$

$$r = 1, 2, \dots, n-1$$

1. Is $\mathbf{H}_{n \times n}$ orthogonal?
2. What is $\mathbf{y}^T \mathbf{h} \mathbf{h}^T \mathbf{y}$ represent?
3. What is $\mathbf{y}^T \mathbf{H}_0^T \mathbf{H}_0 \mathbf{y}$ represent?

1.11.2 Symmetric Matrices

Any quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y} = \mathbf{y}^T \mathbf{B} \mathbf{y}$ where $\mathbf{B} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$ is symmetric. Furthermore, any quadratic form can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y}$ for an infinite number of matrices, but can only be written in one way as $\mathbf{y}^T \mathbf{B} \mathbf{y}$ for $\mathbf{B}$ symmetric. For example,

$$4y_1^2 + 6y_1y_2 + 7y_2^2 = \begin{bmatrix} y_1 & y_2 \end{bmatrix} \begin{bmatrix} 4 & 3+a \\ 3-a & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

for any value of a , but only when $a = 0$ is the matrix involved symmetric. This means that for any particular quadratic form there is only one, unique matrix such that the quadratic form can be written as $\mathbf{y}^T \mathbf{A} \mathbf{y}$ with $\mathbf{A}$ being symmetric. Due to the uniqueness of this symmetric matrix, the quadratic form that we are going to discuss is confined to the case of $\mathbf{A}$ being symmetric.

1.11.3 Positive Definiteness

Definition 27.

A quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ is said to be **positive definite** (p.d.) if

$$\mathbf{y}^T \mathbf{A} \mathbf{y} > 0 \text{ for all } \mathbf{y} \text{ except } \mathbf{y} = \mathbf{0}.$$

The corresponding (symmetric) matrix is also described as positive definite.

Definition 28. A quadratic form $\mathbf{y}^T \mathbf{A} \mathbf{y}$ is said to be **positive semi-definite** (p.s.d) if

$$\mathbf{y}^T \mathbf{A} \mathbf{y} \geq 0 \text{ for all } \mathbf{y} \neq \mathbf{0}$$

with $\mathbf{y}^T \mathbf{A} \mathbf{y} = 0$ for at least one $\mathbf{y} \neq \mathbf{0}$.

The corresponding (symmetric) matrix $\mathbf{A}$ is a p.s.d. matrix.

Example 21. Show that

$$\mathbf{A} = \begin{pmatrix} 3 & 5 & 1 \\ 5 & 13 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

is a positive definite matrix.

Example 22.

Show that

$$\mathbf{B} = \begin{pmatrix} 13 & -2 & -3 \\ -2 & 10 & -6 \\ -3 & -6 & 5 \end{pmatrix}$$

is a positive semidefinite matrix.

1.12 Spectral Decomposition

Result 13. The spectral decomposition of a $k \times k$ symmetric matrix $\mathbf{A}$ with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ (with $\mathbf{u}_i^T \mathbf{u}_i = 1$ and $\mathbf{u}_i^T \mathbf{u}_j = 0$) is

$$\begin{aligned} \mathbf{A} &= \lambda_1 \mathbf{u}_1 \mathbf{u}_1^T + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^T + \dots + \lambda_k \mathbf{u}_k \mathbf{u}_k^T \\ &= \mathbf{U} \mathbf{D} \mathbf{U}^T \end{aligned}$$

where

$$\mathbf{D} = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_k \end{bmatrix}$$

and

$$\mathbf{U} = [\mathbf{u}_1 \mid \mathbf{u}_2 \mid \dots \mid \mathbf{u}_k]$$

is an orthogonal matrix.

Result 14. If $\mathbf{A}$ is a $k \times k$ symmetric nonsingular matrix with spectral decomposition

$$\mathbf{A} = \sum_{i=1}^k \lambda_i \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \mathbf{D} \mathbf{U}^T$$

then

$$(i) \mathbf{A}^{-1} = \sum_{i=1}^k \lambda_i^{-1} \mathbf{u}_i \mathbf{u}_i^T = \mathbf{U} \mathbf{D}^{-1} \mathbf{U}^T$$

(ii) the square root matrix

$$\mathbf{A}^{1/2} = \sum_{i=1}^k \sqrt{\lambda_i} \mathbf{u}_i \mathbf{u}_i^T$$

has the properties:

$$(a) \mathbf{A}^{1/2} \mathbf{A}^{1/2} = \mathbf{A}$$

$$(b) \mathbf{A}^{1/2} \mathbf{A}^{-1} \mathbf{A}^{1/2} = \mathbf{I}$$

$$(c) \mathbf{A}^{1/2} \text{ is symmetric}$$

(iii) The inverse square root matrix

$$\begin{aligned} \mathbf{A}^{-1/2} &= \sum_{i=1}^k \frac{1}{\sqrt{\lambda_i}} \mathbf{u}_i \mathbf{u}_i^T \\ &= \mathbf{U} \mathbf{D}^{-1/2} \mathbf{U}^T \end{aligned}$$

has the properties:

$$(a) \mathbf{A}^{-1/2} \mathbf{A}^{-1/2} = \mathbf{A}^{-1}$$

$$(b) \mathbf{A}^{-1/2} \mathbf{A} \mathbf{A}^{-1/2} = \mathbf{I}$$

$$(c) \mathbf{A}^{-1/2} \text{ is symmetric}$$

In parts (ii) and (iii), $\mathbf{A}$ should be positive definite to ensure that

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0$$

1.13 Random Vectors:

Definition 29.

A random vector $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ is a vector whose elements are random variables.

1.13.1 Mean vectors:

$$E(\mathbf{y}) = \begin{bmatrix} E(y_1) \\ \vdots \\ E(y_n) \end{bmatrix} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = \boldsymbol{\mu}$$

where

$$\mu_i = E(y_i) = \int_{-\infty}^{\infty} y f_i(y) dy$$

if y_i is a continuous random variable with density function $f_i(y)$

and

$$\mu_i = E(y_i) = \sum y p_i(y)$$

if y_i is a discrete random variable with probability function $p_i(y)$.

1.13.2 Covariance matrix:

$$\boldsymbol{\Sigma} = \text{Var}(\mathbf{y}) = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \sigma_{13} & \cdots & \sigma_{1n} \\ \sigma_{21} & \sigma_2^2 & \sigma_{23} & \cdots & \sigma_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_{n1} & \sigma_{n2} & \sigma_{n3} & \cdots & \sigma_n^2 \end{bmatrix}$$

with variances

$$\text{Var}(y_i) = \sigma_i^2 = E(y_i - \mu_i)^2$$

$$= \begin{bmatrix} \int_{-\infty}^{\infty} (y - \mu_i)^2 f_i(y) dy & \text{if } y \text{ is a continuous} \\ & \text{random variable} \\ \sum_{all \ y} (y - \mu_i)^2 p_i(y) & \text{if } y \text{ is a discrete} \\ & \text{random variable} \end{bmatrix}$$

and covariances:

$$\sigma_{ij} = Cov(y_i, y_j) = E[(y_i - \mu_i)(y_j - \mu_j)]$$

where

$$\sigma_{ij} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (y - \mu_i)(v - \mu_j) f_{ij}(y, v) dy dv$$

if y_i and y_j are continuous random variables with joint density function $f_{ij}(y, v)$ and

$$\sigma_{ij} = \sum_{\substack{\text{all} \\ y}} \sum_{\substack{\text{all} \\ v}} (y - \mu_i)(v - \mu_j) P_{ij}(y, v)$$

if y_i and y_j are discrete random variables with joint probability function

$$p_{ij}(y, v) = Pr(y_i = y, V_j = v)$$

Result 15.

Let $\mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ be a random vector with

$$\boldsymbol{\mu} = E(\mathbf{y}) \quad \text{and} \quad \boldsymbol{\Sigma} = Var(\mathbf{y}),$$

and let

$$\mathbf{A}_{p \times n} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{p1} & \cdots & a_{pn} \end{bmatrix}$$

be a matrix of non-random elements, and let

$$\mathbf{c} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad \mathbf{d} = \begin{bmatrix} d_1 \\ \vdots \\ d_p \end{bmatrix}$$

be vectors of non-random elements, then

- (i) $E(\mathbf{A}\mathbf{y} + \mathbf{d}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{d}$
- (ii) $Var(\mathbf{A}\mathbf{y} + \mathbf{d}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T$
- (iii) $E(\mathbf{c}^T \mathbf{y}) = \mathbf{c}^T \boldsymbol{\mu}$
- (iv) $Var(\mathbf{c}^T \mathbf{y}) = \mathbf{c}^T \boldsymbol{\Sigma} \mathbf{c}$

Example 23.

Let the 3×1 random vector $\mathbf{y}$ follows a multi-variate normal distribution with mean vector $\boldsymbol{\mu} = [7 \ 9 \ 5]^T$ and covariance matrix $\boldsymbol{\Sigma}$ where

$$\boldsymbol{\Sigma} = \sigma^2 \begin{bmatrix} 1 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & 3 \end{bmatrix}$$

Consider the vector $\mathbf{w}$ where

$$\mathbf{w} = \begin{bmatrix} 3y_1 - y_2 + 2y_3 - 25 \\ 2y_1 + y_2 - 4y_3 - 12 \end{bmatrix}$$

- (a) Find the mean vector of $\mathbf{w}$.
- (b) Find the covariance matrix of $\mathbf{w}$.